MAE 1001

Introduction to Mechanical and Aerospace Engineering

Course website: https://gwu-mae1001.github.io/fall2025/

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Photo: Kartik Bulusu

What is a Matrix ?

DATA

- Arranged in ROWS and COLUMNS
- Typically carries a MEANING

DATA

- Rectangular ARRAY of numbers

ARRAYS

- Two-dimensional arrays
- *m* rows and *n* columns

Source: http://giphy.com/search/matrix-gif

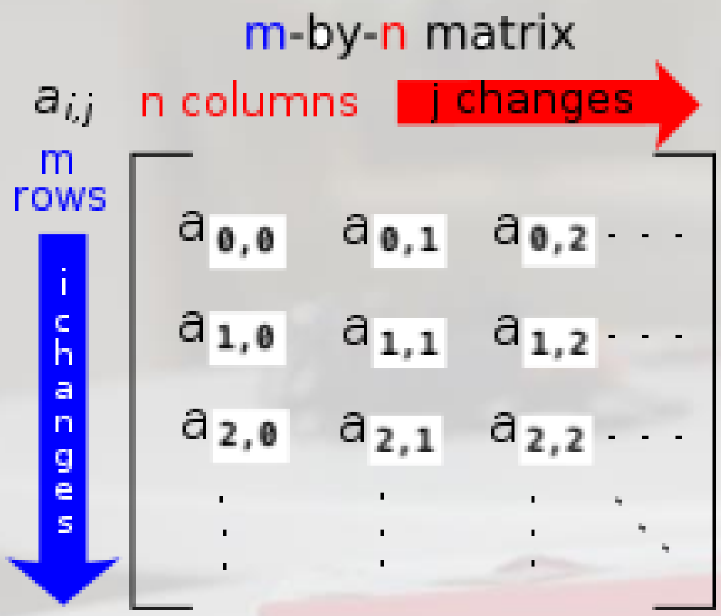
$$\begin{bmatrix} 1 & -4 \\ 9 & 6 \end{bmatrix}$$

$$\begin{bmatrix} 15 & 3 & 9 \\ 2 & 5 & 4 \end{bmatrix}$$

$$\begin{bmatrix} 11 & 7 \\ 4 & 2 \\ 6 & 9 \\ 3 & 1 \end{bmatrix}$$

Photo: Kartik Bulusu

Bookkeeping in a Matrix



Source: http://en.wikipedia.org/wiki/Matrix_(mathematics)

Python:

```
>>> import numpy as np
>>> A = np.matrix([[-1, 2],[3, 4]])
>>> A[0,0]
>>> A[0,:]
>>> A[:,0]
>>> A[1,0]
```

MATLAB:

```
>> A = [-1 2; 3 4]
>> A(1,1)
>> A(1,:)
>> A(:,2)
>> A(2,1)
```

The ORDER of a matrix

- $A_{m\times n}$ *is* $m \times n$
- *Read as "m-by-n"*

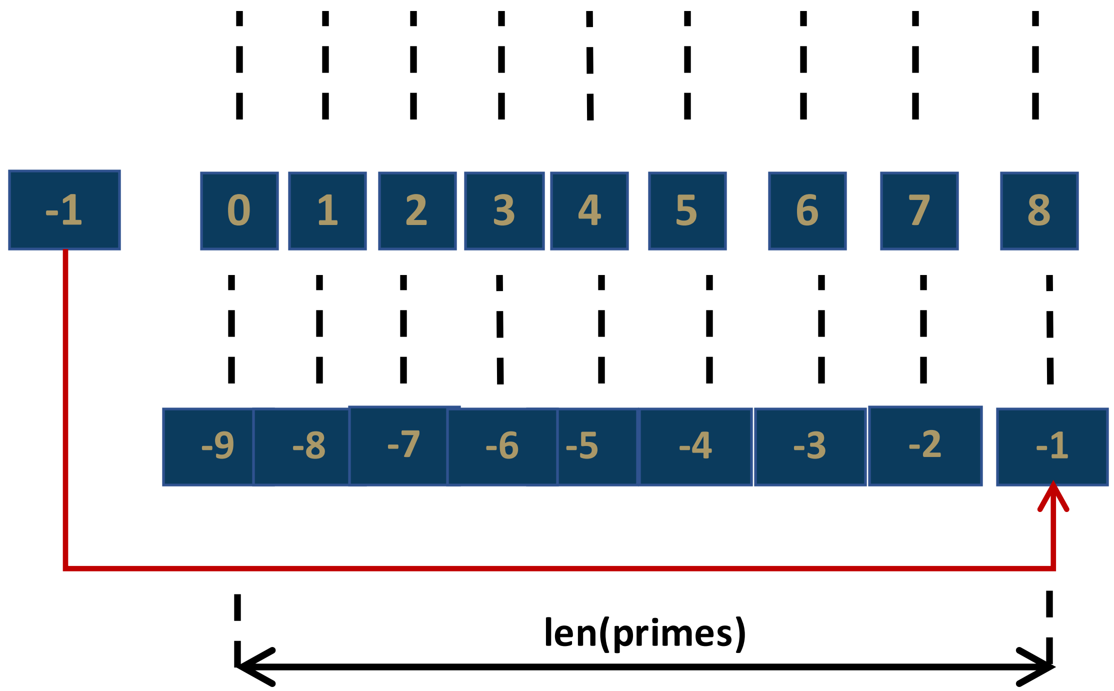
a_{ij} *is called an ELEMENT*

- *at the i^th row and j^th column of A*

Indexing and Slicing Lists

Retrieve list-elements with a range of values

```
>>> primes = [2, 3, 5, 7, 9, 11, 13, 17, 19]
```



start *stop*

```
>>> primes[2:5]
[5, 7, 9]
```

start *stop* *step*

```
>>> primes[0:7:2]
[2, 5, 9, 13]
```

start *stop* *step*

```
>>> primes[8:2:-2]
[19, 13, 9]
```

start:	at the index value
step:	up or down at the increment value (default = 1)
stop:	at the index value but not including it

Example of who is using the sense HAT and where - Astro Pi

Source: https://youtu.be/kk_7KNuRLrk

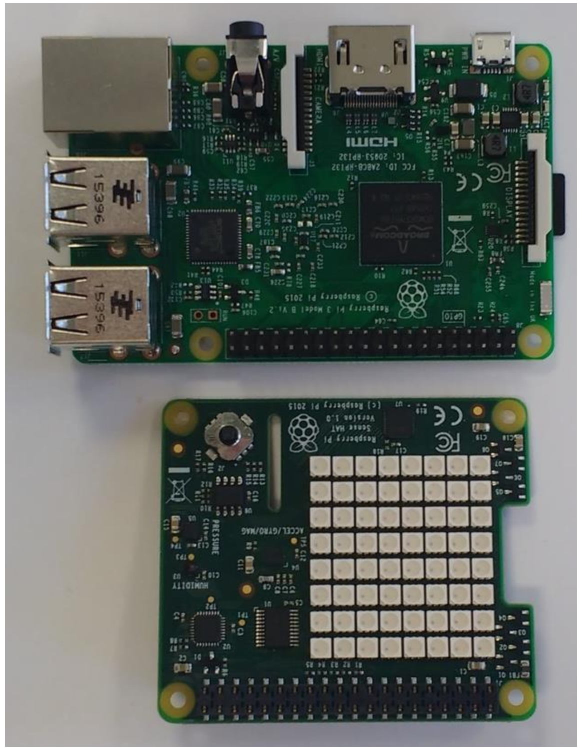
What we will do today

- Co-work
  - Observe, ask and try in groups
- Write small program using Python
- Think about
  - Challenges, Opportunities, Gaps and Surprises

What we will learn today

- Communicate with the Sense HAT using Python
- Access the outputs of the Sense HAT
- Use the Sense HAT library to display messages and images
- Use loops to repeat certain code blocks

Step-1: “Put on the Sense HAT”



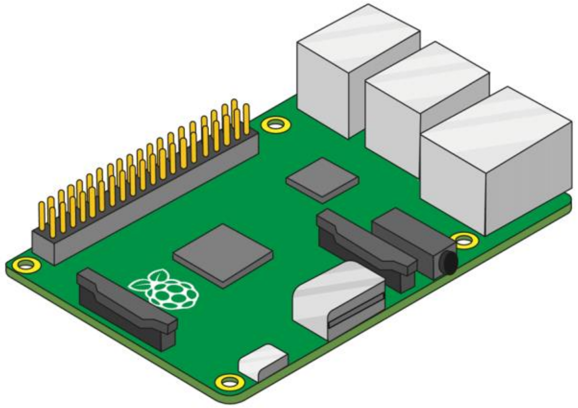


Image and animation source: https://projects.raspberrypi.org/en/projects/getting-started-with-the-sense-hat/2

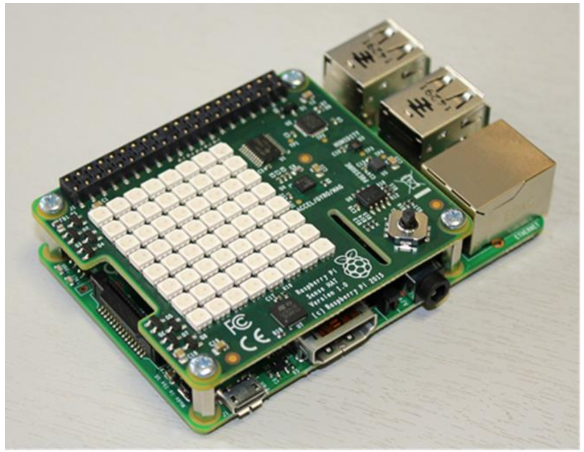


Image source: https://reference.wolfram.com/language/ref/device/SenseHAT.html

Source: https://youtu.be/8NwWNOMqai4

“and take a closer look” …

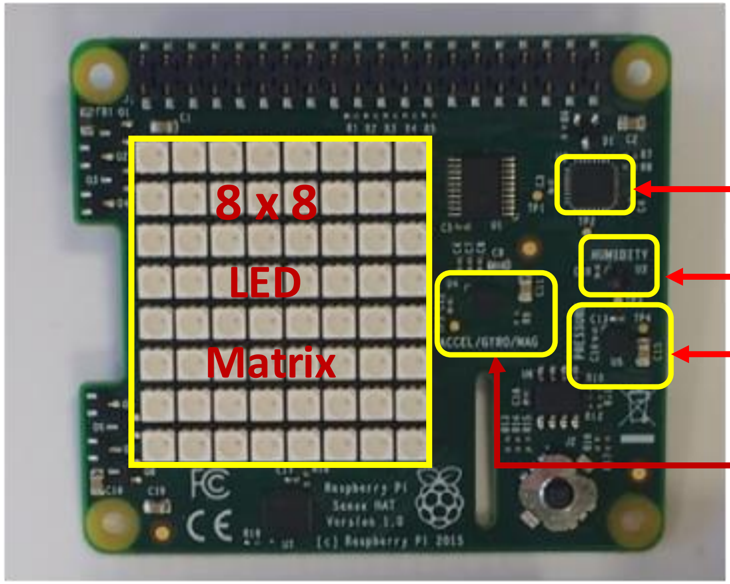
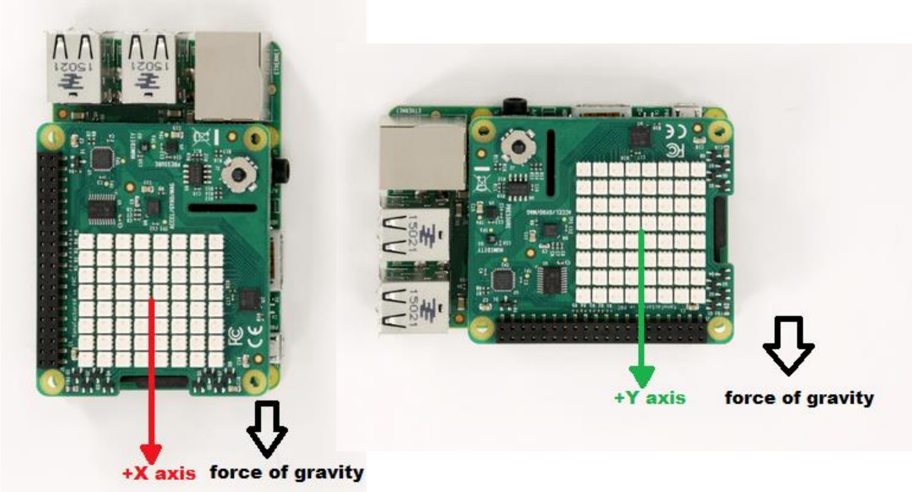


Image source: https://projects.raspberrypi.org/en/projects/getting-started-with-the-sense-hat/2

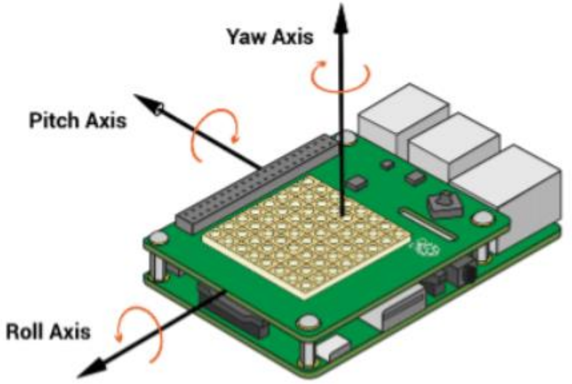
- The Sense HAT has a variety of sensors that can be read from:

"Temperature"	reads temperature in degrees Celsius
"Humidity"	reads humidity in % RH
"Pressure"	reads atmospheric pressure in millibars
"Rotation"	reads gyroscopic motion in revolutions per second
"Acceleration"	reads acceleration in terms of standard accelerations due to gravity on Earth's surface
"Orientation"	reads orientation relative to magnetic north in degrees
"Magnetic Field"	reads strength and direction of a magnetic field around the sensor in microteslas

- The gyroscope, accelerometer, and magnetometer sensors return a list of three values that corresponds to $\{roll, pitch, yaw\}$, as oriented according to the following image:



Source: https://www.mathworks.com/help/supportpkg/raspberrypi/examples/auto-rotate-an-image-displayed-on-sense-hat-led-matrix.html



Source: https://reference.wolfram.com/language/ref/device/SenseHAT.html

Starting point for further exploration:
[Link for "Getting started with the Sense HAT"]